Lost & Found Network

Requirements Gathering Report

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1. Project Overview and Context

1.1 Project Summary

The Lost & Found Network is a web-based application designed to simplify the process of reporting, searching for, and claiming lost and found items within the George Brown Polytechnic campuses. The system enables students, staff, and campus visitors to submit reports for items they have lost or found.

1.2 Project Context

George Brown Polytechnic operates across multiple campus locations, each with various buildings, classrooms, common areas, and facilities where items are frequently misplaced or left behind. Currently, lost and found services are fragmented; items may be turned into security desks, departmental offices, or library service points, with no unified system for tracking or publicizing found items. Individuals who lose belongings often do not know where to inquire, leading to unclaimed items and frustrated community members.

The Lost & Found Network will serve as a centralized digital platform accessible to the college community. Primary users include students who may lose personal items such as phones, wallets, or textbooks; staff members who encounter or lose items in office and shared spaces; and campus visitors attending events or using facilities like the library or gym.

2. Stakeholder Analysis

2.1 Stakeholder List

Stakeholders include students, faculty and staff, lost and found office/ security staff, IT department, and college administration.

Stakeholder	Role	Interests / Goals	Influence
Students	Primary users who lose or find items on campus	Quickly report lost items, search for found items, securely recover belongings	High
Faculty and Staff	Users and occasional administrators of lost items	Efficient handling of lost property, reduced manual work, fewer inquiries	High

Campus Visitors	Occasional users (guests, event attendees)	Simple way to report and recover lost belongings	Low
Lost & Found Office / Security Staff	Operational managers of the process	Accurate tracking, dispute resolution, secure item handover	High
IT Department	System maintenance and support	System reliability, security, scalability	Medium
College Administration	Institutional oversight	Improved service quality, reduced complaints, accountability	Medium

2.2 Stakeholder Summary

The most important stakeholders for the Lost & Found Network for Campus are students, faculty and staff, and Lost & Found or Security staff. These people will use the system directly, so they are affected the most. Students and visitors mostly just want to find their lost items quickly and use the system easily without confusion. Faculty and staff also want the process to be faster. They don't have to keep answering the same lost item questions again all the time.

Lost & Found and Security staff have high influence because they are the ones who check ownership, deal with disputes, and return items. Sometimes there can be problems because users tend to claim items fast, but staff still need to follow rules and check details carefully to avoid any conflict such as giving items to the wrong person.

The IT department and college administration are not using the system every day, but they are responsible for security, system reliability, and overall service quality. For this system, it is important to balance easy use for users and proper control for staff.

3. Business Need and Problem Statement

3.1 Business Need

In George Brown Polytechnic and other big campuses, when someone loses something, the process of finding it back is very confusing. Different buildings have their own lost and found desks, some use paper, some just keep items in a box. Students and staff usually don't know where to go first, so they walk around or ask many offices. Many

times, the person who found the item also doesn't know where to leave it. Those who find items may not know the proper place or procedure to turn them in. This fragmented approach results in confusion, wasted time, and a poor experience for the campus community.

Because of this, many items stay unclaimed for a long time, and people get frustrated. Staff also spend a lot of time answering same questions repeatedly, like "did someone bring my wallet?" or "is there a phone found in the library?". This makes the work slower and not organized. A better way is needed to collect all information in one place so everyone can check easily.

3.2 Problem Statement

There is no unified system at George Brown College that allows students, staff, and visitors to report lost items, register found items, and verify ownership through a single, consistent process. Instead, information is dispersed across separate offices and informal channels, resulting in limited discoverability, delayed responses, and frequent disputes over ownership. This lack of centralization causes unnecessary effort for users, inefficient handling by staff, and a low success rate in returning lost property to its rightful owners.

4. Project Goals and Scope

4.1 High-Level Goals

1. **Provide a centralized lost and found system**

Create a system that would enable students, staff and visitors to report, search and claim lost or found items

2. **Improve the efficiency and success chance of item recovery**

Increase item recovery chance by having detailed reporting, listings that can be searched and claim verification

3. **Ensure secure item claiming**

Create a claim process that would allow found item reporters to verify ownership before giving the item

4. **Provide updates to the user**

Notify the user when a potential match happens to increase the chance of a quick recovery

5. **Support administrative oversight and dispute resolution**

Provide tools that manage listings, resolve disputes, and manage data for the administrators

4.2 Scope – In Scope

1. **User accounts and authentication**
    a. Account creation using email
    b. Secure login and logout
2. **Lost item reporting**
    a. User creation of lost item reports with category, description, date lost, location, and optional image
    b. Storage and display of reports in lost item listings
    c. System checking for potential matches
3. **Found Item Reporting**
    a. Users' creation of found item reports with category, description, date found, location, and image
    b. Storage and display of found item listings
    c. System checking for potential matches
4. **Search and Browse Functionality**
    a. Users can browse lost and found listings
    b. Users can filter listings by category, date time, location, and status
    c. Users can view detailed information for listings
5. **Claim Process**
    a. Users can submit claims for found items
    b. Admins can review, approve, or reject claims
    c. Upon approval, the item becomes available for collection
6. **Administrative Management**
    a. Admin dashboard for viewing and managing all listings
    b. Ability to edit or remove inappropriate listings
    c. Management of item categories
    d. Resolution of disputes
    e. Logging of administrative actions
7. **Email Notifications**
    a. Email notifications are sent when potential matches are detected between lost and found reports or when a claim has been approved

4.3 Scope – Out of Scope

- Mobile application, system will be web-based only
- Real-time chat or direct messaging between users
- Image recognition for lost and found items
- SMS notifications
- Physical item storage tracking
- Advanced analytics dashboards
- Public API access
- Automatic content moderation

5. Functional Requirements

FR-1: The system shall allow users to create an account using any valid email address.

FR-2: The system shall allow authenticated users to securely log in and log out.

FR-3: The system shall allow authenticated users to report a lost item by providing category, description, date lost, last known location, and optional reference image.

FR-4: The system shall save lost item reports and display them in the lost items listings with status indicators (active, claimed, expired).

FR-5: The system shall allow users to edit or delete their own lost item reports.

FR-6: The system shall allow users to browse all active lost and found item listings.

FR-7: The system shall allow users to filter listings by category, date range, campus location, and item status.

FR-8: The system shall allow users to submit a claim for a found item by providing verification details (identifying information only the owner would know).

FR-9: The system shall notify the found item reporter when a claim is submitted for their listing.

FR-10: The system shall allow found item reporters to review claim submissions and approve or reject them with optional feedback.

FR-11: The system shall mark items as "claimed" and update listing status when a claim is approved.

FR-12: The system shall provide authenticated admin users with the ability to view all listings regardless of status.

FR-13: The system shall allow admin users to edit or remove listings that violate policies or contain inappropriate content.

6. Non-Functional Requirements

NFR-1 (Usability): The system should offer feature parity between the mobile and desktop web interfaces with effective scalability.

NFR-2 (Privacy): The system should limit the visible data regarding users who post lost items, only showing the information needed to facilitate the return of lost items.

NFR-3 (Security): The system should authenticate users following modern best practices for storing user data.

NFR-4 (Accessibility): The system should be easy to use for people who are visually impaired, utilizing a color palette that doesn't negatively affect users who are colorblind, as well as optimizing the HTML for use with screen readers.

NFR-5 (Reliability): The system should have minimal downtime during each semester while the campus is more active with in-person students.

7. Constraints and Assumptions

7.1 Constraints

1. **Privacy and Data Protection Constraint**

The system must limit the collection, storage, and public display of personal information

2. **Resource Constraint**

The project is limited to the tools and technologies available to a student development team.

3. **Time Constraint**

The system must be analyzed and documented within the timeframe of a single academic semester.

4. **Access Control Constraint**

Access to the system is limited to individuals with an established relationship to the school, such as students, staff, or authorized visitors.

5. **Operational Scope Constraint**

The system is limited to managing digital records related to lost and found items, while the physical storage, handling, and return of items are managed by campus staff.

7.2 Assumptions

1. **User Access Assumption**

It is assumed that users have access to the internet to use the system.

2. **User Reporting Assumption**

It is assumed that users will provide accurate and truthful information when submitting lost or found item reports.

3. **Responsible Use Assumption**

It is assumed that users will not intentionally misuse the system to falsely claim items that do not belong to them.

4. **Administrative Oversight Assumption**

It is assumed that designated campus staff will use the system according to established procedures when managing, storing, and returning lost items.

5. **System Adoption Assumption**

It is assumed that students will choose to use the centralized lost and found system rather than relying on informal or alternative methods.

8. Initial Use Cases and/or User Stories

Use Case 1: Report a lost item

- Actor: Student, Staff Member, Campus Visitor
- Precondition: User is authenticated and logged into the system
- Main Flow
    1. User selects "Report Lost Item" from the dashboard.
    2. User enters item details including category, description, date lost, and last known location.
    3. User optionally uploads a reference image.

4. submits the lost item report.
5. System saves the report and displays confirmation.
6. System checks for potential matches with existing found item reports.

- Outcome:
  Lost item report is created and visible in the system's lost items listings.

Use Case 2: Report a Found item

- Actor: Student, Staff Member, Campus Visitor
- Precondition: User is authenticated and logged into the system
- Main flow:
  7. User selects "Report Found Item" from the dashboard.
  8. enters item details including category, description, date found, and location where item is being held.
  9. User uploads an image of found item
  10. submits the found item report.
  11. System saves the report and displays confirmation.
  12. checks for potential matches with existing lost item reports.
- Outcome:
  1. Found item report is created and visible in the system's found items listings.
  2. If a potential match exists, system notifies the lost item reporter via email.

Use Case 3: Search and Browse Listings

- Actor: Student, Staff Member, Campus Visitor
- Precondition: Listing is visible
- Main flow:
  1. User navigates to the search/browse page.
  2. User optionally applies filters by category, date range, location, or item status.
  3. User enters search keywords if desired.
  4. System displays matching listings sorted by date.
  5. User selects a listing to view full details.
- Outcome:
  User views relevant lost or found item listings matching their search criteria.

Use Case 3: Claim a Found Item

- Actor: Student, Staff Member, Campus Visitor (as claimant); Found Item Reporter (as verifier)
- Precondition: User is authenticated; found item listing exists and is unclaimed
- Main flow:
  1. Claimant locates a found item listing that appears to be theirs.
  2. Claimant selects "Claim This Item" and provides verification details (specific identifying information only the owner would know).
  3. System notifies the found item reporter of the claim request.

    4. Found item reporter reviews the claim and verification details.
    5. Reporter approves or rejects the claim.
    6. If approved, system marks item as claimed and facilitates exchange details.
- Outcome:
    - Item is marked as claimed, and both parties receive confirmation with next steps for retrieval.
    - If reporter rejects the claim, claimant is notified and may submit additional verification. If multiple claims exist, reporter reviews each and may escalate to admin for dispute resolution.

Use Case 4: Manage Listings (Admin)

- Actor: Staff Member (System Administrator)
- Precondition: Admin is authenticated with administrative privileges
- Main flow:
    1. Admin logs into the admin dashboard.
    2. Admin views all active listings with filtering options.
    3. Admin selects a listing to review.
    4. Admin can edit listing details, remove inappropriate content, or delete outdated listings.
    5. System logs the administrative action.
    6. System updates the listing status accordingly.
- Outcome:
    - If listing is involved in an active dispute, system warns admin before deletion. Admin may archive rather than delete for record-keeping.
    - Listing is updated, removed, or archived based on administrative action.

Use Case 5: Resolve Claim Dispute (Admin)

- Actor: Staff Member (System Administrator)
- Precondition: Admin is authenticated with administrative privileges and A dispute has been escalated involving multiple claimants or a rejected claim appeal
- Main flow:
    1. Admin receives notification of escalated dispute.
    2. Admin reviews the found item details, all claim submissions, and verification information provided.
    3. Admin contacts involved parties if additional information is needed.
    4. Admin makes a determination and assigns the item to the verified owner or marks as unresolved.
    5. System notifies all parties of the decision.
    6. If ownership cannot be determined, admin may require in-person verification with ID and proof of ownership.
- Outcome:
  Dispute is resolved, item is assigned to rightful owner or flagged for in-person

resolution.

User Stories
User can be a student, staff member or campus visitor

1. As a user, I want to report a lost item with details and an image so that others can help identify and return it to me.
2. As a user, I want to report a found item so that the rightful owner can locate and claim it.
3. As a user, I want to search found items by category and location so that I can quickly find my lost belongings.
4. As a person who found an item, I want to verify a claimant's identity by asking specific questions so that I can ensure the item goes to the rightful owner.
5. As a user, I want to receive an email notification when a matching found item is posted so that I can claim it quickly.
6. As an administrator, I want to remove old or inappropriate listings so that the system remains useful and current.
7. As an administrator, I want to resolve disputes between multiple claimants so that items are returned to their rightful owners fairly.

9. Priorities and Risk Highlights

9.1 Requirement Priorities

To help manage scope and focus development effort, key functional requirements have been prioritized using the following categories:

- **Must Have** – Essential for the system to be considered functional
- **Should Have** – Important but not strictly required for the initial release
- **Could Have** – Nice-to-have features that may be implemented if time permits

Must Have Requirements

These requirements represent the core functionality without which the system cannot fulfill its primary purpose.

- **FR-1:** User account creation using email address (institutional emails receive "Verified GBC Member" badge)
- **FR-2:** Secure user login and logout

- **FR-3:** Report a lost item with category, description, date, location, and optional image
- **FR-4:** Report a found item with category, description, date, location, and image
- **FR-5:** Browse and search lost/found listings with filters
- **FR-6:** Submit a claim for a found item with verification details
- **FR-7:** Found item reporters can review, approve, or reject claims
- **FR-8:** Contact information revealed to both parties upon claim approval

Should Have Requirements

These requirements add significant value and improve user experience but are not strictly essential for a minimum viable product.

- **FR-9:** Email notifications when a potential match is detected between lost and found reports
- **FR-10:** Admin management of item categories (create, edit, delete)
- **FR-11:** Admin removal of outdated or inappropriate listings
- **FR-12:** Admin dispute resolution when multiple users claim the same item
- **FR-13:** Users can edit or delete their own reports
- **FR-14:** Display report status (active, claimed, expired) to users

Could Have Requirements

These requirements represent enhancements that would improve the system but can be deferred to future development if time does not permit.

- **FR-15:** Automatic archival of listings after a defined retention period (e.g., 90 days)
- **FR-16:** User dashboard showing submitted reports and claim history
- **FR-17:** Option to mark a lost item as "found elsewhere" and close the report
- **FR-18:** Support for multiple images per report
- **FR-19:** Basic admin analytics (e.g., number of reports, claim success rate)

Non-Functional Requirement Priorities

Priority	Requirement
Must Have	NFR-1 (Security): Secure storage of user authentication data using industry-standard practices
Must Have	NFR-2 (Usability): Accessible on modern desktop and mobile browsers
Should Have	NFR-3 (Performance): Listings and search results load within 3 seconds under normal usage
Should Have	NFR-4 (Availability): 99% uptime during the academic semester
Could Have	NFR-5 (Accessibility): Compliance with WCAG 2.1 Level AA guidelines

9.2 Easly Risk Highlights

Several early risks have been identified based on the current requirements and project context:

Scope Creep Risk

Additional features such as direct messaging, mobile applications, or integration with campus security systems may be requested during development.

Mitigation: Clearly define in-scope and out-of-scope items and refer back to these boundaries when evaluating new feature requests. Any additions must be offset by removing or deferring existing requirements.

Technical Complexity Risk

The notification system for matching lost and found items requires logic to compare descriptions, categories, dates, and locations. Effective matching without excessive false positives may prove challenging.

Mitigation: Start with simple matching criteria (same category, overlapping date range, same campus location) and refine based on testing. Defer advanced matching (e.g., keyword analysis) to “Could Have” status.

Claim Verification Risk

The structured verification process relies on found item reporters asking appropriate questions. Weak or easily guessed questions could enable fraudulent claims.

Mitigation: Provide guidance and example verification questions during report submission. Allow administrators to intervene in disputed claims. Non-institutional users accept reduced trust status. Found item reporters may prefer to approve claims from verified GBC members.

Team Experience Risk

Team members may have varying experience levels with the chosen technology stack, particularly around image upload handling, email integration, and secure authentication.

Mitigation: Select technologies familiar to at least one team member. Assign tasks based on individual strengths. Identify learning requirements early and allocate time for knowledge sharing.

Time Constraint Risk

The project must be completed within a single academic semester, and the five-person team size further constrains capacity.

Mitigation: Prioritize Must Have requirements first and develop in small iterations. Conduct weekly progress reviews to identify delays early. Be prepared to move “Should Have” items to “Could Have” if timeline pressures emerge.

Data Privacy Risk

The system will store user contact information and item descriptions, some of which may inadvertently contain sensitive details.

Mitigation: Collect only necessary information. Reveal contact details only upon successful claim approval. Include privacy notice during registration. Ensure secure data storage and transmission practices.

Low Adoption Risk

If the campus community is unaware of the system or finds it difficult to use, adoption may be low, reducing platform effectiveness.

Mitigation: Design an intuitive user interface requiring minimal training. Gather early user feedback to address usability issues.

Risk Summary Table

Risk	Likelihood	Impact	Mitigation Strategy
Scope Creep	Medium	High	Strict scope boundaries, change control
Technical Complexity (Matching)	Medium	Medium	Start simple, iterate based on testing
Claim Verification Weakness	Low	Medium	User guidance, admin oversight
Team Experience Gaps	Medium	Medium	Task assignment by strength, knowledge sharing
Time Constraints	High	High	Prioritized backlog, weekly reviews
Data Privacy Concerns	Low	High	Minimal data collection, secure practices
Low Adoption	Medium	Medium	Intuitive UI, user feedback loops

10. Summary and Next Steps

In this report, we have outlined the critical details regarding the project's overview and context, established key stakeholders involved, and determined the primary business needs and problem statement. We have identified the necessary requirements, both functional and non-functional as well as key constraints and assumptions. We have drafted use cases and user stories to start ourselves off with and considered the main priorities and risk highlights. With the information contained we have set ourselves up with a valuable framework which we will utilize as we progress through the development of this project.

In the upcoming phases, we will build upon what we have defined here by creating an in-depth project proposal. This will allow us to establish a more focused business need and project scope and progress to the stage of identifying the elements of our technology stack. We will review the feasibility, constraints and risks of our requirements to then generate a structured project plan and schedule. We will also determine what our individual roles will be and how we plan to communication and collaborate with one another to make our work as efficient as possible. The information in our requirements gathering report will provide us with an effective launch point as we continue our progress towards the project goals.